

Roll No.

TPE-611

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
ALTERNATE ENERGY RESOURCES**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss in details about the energy sources and their availability with respect to commercial or conventional energy resources. (CO1)

OR

- (b) What do you mean by fluidized bed technology for the coal gasification ? Discuss the principle of fluidization and fluidized bed combustion. (CO1)

2. (a) What are the renewable energy resources ? Discuss advantages and prospects of renewable energy resources. (CO2)

OR

- (b) Define wind energy ? What are the different types of wind mills designed and developed ? Discuss in detail. (CO2)

P. T. O.

(2)

3. (a) Define the following : (CO2)
- (i) Solar constant
 - (ii) Solar radiation
 - (iii) Diffuse radiation
 - (iv) Airmass

OR

- (b) Discuss the solar radiation geometry specifying the different angle used in energy calculations with the help of diagrams. (CO2)
4. (a) What do you mean by Gas hydrates ? Explain some features of the gas hydrates ? How gas hydrates formed ? Discuss the various issues related to extraction of natural gas hydrates. (CO1, CO2)

OR

- (b) What are the different measuring devices for solar radiation ? Discuss one of them briefly. (CO1, CO2)
5. (a) Define collection system, characteristic features of collector system for the solar energy. (CO2)

OR

- (b) Discuss in details about the types of collectors ? Explain the performance analysis of flat plate collectors (FPC). (CO2)

Roll No.

TPE-605

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
PETROLEUM PRODUCTION ENGINEERING—II**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What are the different types of loads on offshore structures ? Explain any *four*. (CO1)

OR

- (b) With the help of a diagram, describe perforation and explain shaped charge perforation operations. How is corrosion protection done in offshore platforms and how oil is transported from offshore oil and gas facility ? (CO1)

2. (a) Describe tension leg platforms, concrete gravity structures and compliant towers. (CO2)

OR

- (b) Explain production riser. Describe steel catenary riser and well completion. (CO2)

P. T. O.

(2)

3. (a) Write short notes on the following :

(CO1)

- (i) TTR
- (ii) FPSO
- (iii) SPAR Platform

OR

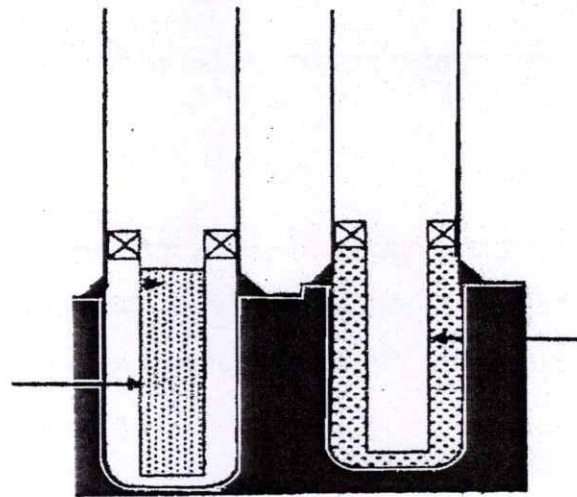
(b) Describe subsea production system and explain all the equipments used for Subsea production. (CO1)

4. (a) Explain, what is well activation and how compressed air can be used for well activation. (CO2)

OR

(b) Describe the methods which are used for removal of salts from sea water. (CO1, CO3)

5. (a) From the given figure, identify types of liner and differentiate between them. (CO1, CO3)



OR

(b) Define oil platforms. Name the different types of fixed and floating platforms.

Roll No

MID-SEMESTER EXAMINATION APRIL 2024

Name of the Program: B. Tech (PE)

Semester: VI

Name of the Course: Data Science

Course Code: TPS 602

Time: 90 Minutes

Note:

- (i) Answer **all the questions** by choosing **any one of the sub-questions**.
(ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	State significant applications of data science in the context of societal benefit. How data science is complementary to AI.	CO1
OR		
(b)	You are provided with a dataset named sales_data.csv, which contains information about sales transactions. The dataset includes the following columns: transaction_id, product_id, customer_id, amount, and date. Your task is to write SQL queries to perform the following tasks: - Retrieve all transactions where the amount is greater than \$100. - Retrieve transactions made by customers with IDs between 1000 and 2000. - Retrieve transactions for a specific product, where the product ID is 'P1234'. - Retrieve transactions made in June 2023. Write SQL queries for each task using the WHERE clause to filter the data accordingly.	
Q2	(10 marks)	
(a)	Discuss various aspects of analytics methodology in data science.	CO1
OR		
(b)	Define the outer joins and all its types. Also, define inner, natural joins with a suitable example. How the data is sorted by ORDER BY clause in a given table.	
Q3	(10 marks)	
(a)	How does an aggregate function work in the database? Clear state the query and its result with a suitable example.	CO2
OR		
(b)	How does data science contribute to the development and advancement of artificial intelligence (AI)? Provide examples of how data science techniques and methodologies are utilized in AI research and applications.	
Q4	(10 marks)	
(a)	Discuss the challenges associated with handling missing data in datasets. Describe three strategies for handling missing data and provide examples of situations where each strategy would be appropriate.	CO2
OR		
(b)	What is the significance of statistics in data science? Consider the following set of numbers representing the scores of students in a math test: 85, 92, 76, 85, 90, 85, 88, 92, 85, 76. Calculate the mean, mode, and median of the scores.	
Q5	(10 marks)	
(a)	Consider a table named customers with the following columns: customer_id, name, age, and city. a) Write an SQL query to retrieve the names of all customers. b) Write an SQL query to retrieve the names and ages of customers residing in the city of 'New York'. c) Write an SQL query to calculate the average age of all customers.	CO2
OR		
(b)	Discuss the importance of interpretability in machine learning models. Describe two techniques for improving the interpretability of machine learning models and explain why interpretability is crucial in real-world applications.	

Roll No.

TPE-601

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)**

MID SEMESTER EXAMINATION, March, 2024

RESERVOIR MODELING AND SIMULATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is the physical properties of rocks ? Write and explain *five* physical properties of rock. (CO1)

OR

- (b) Explain Sedimentary rock. How many types of sedimentary rocks are available on the earth ? Write their relative percentage on the earth.

(CO1)

2. (a) Draw a cross-section of a petroleum system and explain its all important components. (CO1)

OR

- (b) Explain generation, migration and trapping of hydrocarbons in a petroleum system. (CO1)

P. T. O.

(2)

3. (a) Explain Reservoir Engineering and various steps of Reservoir Engineering. What is the difference between Reservoir Modeling and Reservoir Engineering ? (CO2)

OR

- (b) Explain natural and artificial recovery mechanisms in detail. How many types of recovery mechanism are available in oil industry ? (CO2)
4. (a) Explain trap. Illustrate different types of traps by drawing a neat figure. Why are they important ? (CO2)

OR

- (b) Explain Reservoir pressure, Rock texture, Porosity, Permeability and Saturation. (CO2)
5. (a) How Darcy equation is used to measure permeability for flowing fluid and gas system ? (CO2)

OR

- (b) Explain sequential workflow. Draw a sequential reservoir modeling workflows for building a reservoir model. Explain its each component. (CO2)

Roll No.

TPE-603

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)**

MID SEMESTER EXAMINATION, March, 2024

NATURAL GAS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define Natural Gas. Draw and explain the PT and PV phase diagram for single component system showing the different regions. Define bubble and dew point in reservoir of natural gas. (CO1)

OR

- (b) Discuss in detail about the properties of natural gas. (CO1)

Solve the numerical :

A dry gas well is producing a gas stream of the following molar composition : 95 percent methane and 5 percent carbon dioxide. The molar mass of methane is 16 g/mol and that of carbon dioxide is 44 g/mol. Assuming ideal gas behaviour, gas constant $R = 8.31 \text{ J/mol K}$. Find the gas stream density at 10^7 Pa and 350 K in SI unit.

P. T. O.

2. (a) A gas well producing with the following composition : (CO₂)

Composition	Mole Fractions
C1	0.85
C2	0.03
C3	0.025
CO ₂	0.06

Assuming ideal gas behaviour, calculate the following :

- (i) Apparent molecular weight of the gas
- (ii) Gas specific gravity
- (iii) Gas density at 2000 psi and 150 degree F
- (iv) Specific volume at 2000 psi and 150 degree F

OR

- (b) Define the critical and pseudo temperature and pressure of the natural gas reservoir and hence calculate the gas specific gravity of natural gas in the reservoir having the following composition : (CO₂)

Composition	Mole Fractions
C1	0.77
C2	0.08
C3	0.02
CO ₂	0.03
H ₂ S	0.02

(3)

The reservoir is having at initial condition of $P_i = 4000$ psia and 190°F . By calculating sp. gr. of the gas, also calculate T_{pc} , P_{pc} , T_{pr} , P_{pr} by using Standing and Katz correlations.

3. (a) Discuss the stability regions of methane hydrate and carbon dioxide hydrate on the PT curve by drawing the CH_4 hydrate equilibrium curve and CO_2 hydrate equilibrium curve. (CO2)

OR

- (b) Discuss in detail about the coal bed methane. By which phenomena CBM is trapped in the bed seam? Compare Coal Bed Methane (CBM) and conventional reservoir on the different aspects. (CO2)

4. (a) What do you mean by gas hydrates? Explain some features of the gas hydrates. How does gas hydrate formed? Discuss the various issues related to extraction of natural gas hydrates. (CO1, CO2)

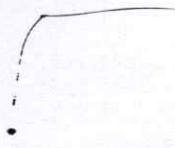
OR

- (b) Gas hydrates extraction requires the unconventional approach, why? Explain the different methods in the extraction of gas hydrate. (CO1, CO2)

5. (a) Explain hydraulic fracturing. What are the theory and techniques for the hydraulic fracturing? Why hydraulic fracturing needed in the production of shale gas? (CO2)

OR

- (b) Describe the steps involved in the extraction of shale gas. Why hydraulic fracturing needed in the production of shale gas? (CO2)



Roll No.

TPE-604

**B. TECH. (PETROLEUM ENGINEERING)
(SIXTH SEMESTER)**

MID SEMESTER EXAMINATION, March, 2024

PETROLEUM DRILLING ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Define directional drilling and deviation control. What are the situations in which directional drilling can be applied ? (CO1)

OR

- (b) When planning directional wells in the oil and gas industry, several parameters need to be carefully considered to ensure successful drilling and production. Describe what are those parameters to be considered in directional well planning. (CO1)

2. (a) Describe what is meant by direction well profile ? List and describe the parameters to be used for defining well profile. Explain different types of well profile in directional drilling with help of a neat diagram. (CO1)

P. T. O.

(2)

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OR

- (b) The following data refer to a directionally drilled well : (CO2)

TVD of station 1 = 1150 ft

Northing coordinate of station 1 = 350 ft

Easting coordinate of station 1 = 550 ft

Target bearing = 65 degree

Survey data at two stations are as follows :

Station	Measured Depth (ft)	Inclination (I)	Corrected Azimuth (A)	Vertical section (ft)
1	1200	15	45	646.3
2	1400	19	55	

Use the radius of curvature method to calculate the well path between stations 1 and 2 and dog-leg severity.

3. (a) Derive the expression for the parameters to be required for geometrical planning of Type I well profile i.e. build up and hold trajectory. (CO2)

OR

- (b) Determine the azimuth with respect to true north of the following wells :

(CO2)

Well	Observed azimuth with respect to magnetic north	Declination
1	N45°E	3° west
2	N45°E	3° east
3	S80°W	5° west

(3)

4. (a) Explain the principle of operation of mud motors in directional drilling.

(CO1)

OR

- (b) What is whipstock ? What role does whipstock technology play in drilling techniques ?

(CO1)

5. (a) The Bottom Hole Assembly (BHA) plays a crucial role in directional drilling by providing the necessary tools and equipment to control the trajectory of the wellbore and achieve specific drilling objectives. Describe how the BHA is used in directional drilling.

(CO1)

OR

- (b) A packed BHA was run from 2500 to 4900 ft. The azimuth over this interval changed from 56° to 65° . The direction to the target from the present BHA location is 58° . If the well is lined up to target on the vertical section (inclination angle is constant) and the estimated MD of the target is 9500 ft, calculate :

(CO2)

- (i) current walk rate
- (ii) total walk needed to hit the target
- (iii) rate of walk required
- (iv) expected walk with the packed BHA

